Homework 9

EE 999 — Introduction to Control Systems

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Due Wednesday, November 18, 2026

Problem 1 — System Identification with `invfreqs()`

Run `HW09_InvfreqsExample.m`. *It fits two transfer functions to simulated frequency-response data: 2 poles, 0 zeros; and 3 poles, 1 zero.*

Problem 1.A

Considering the frequency response plots, which transfer function better fits the data? (The residual appears in the plot title.)

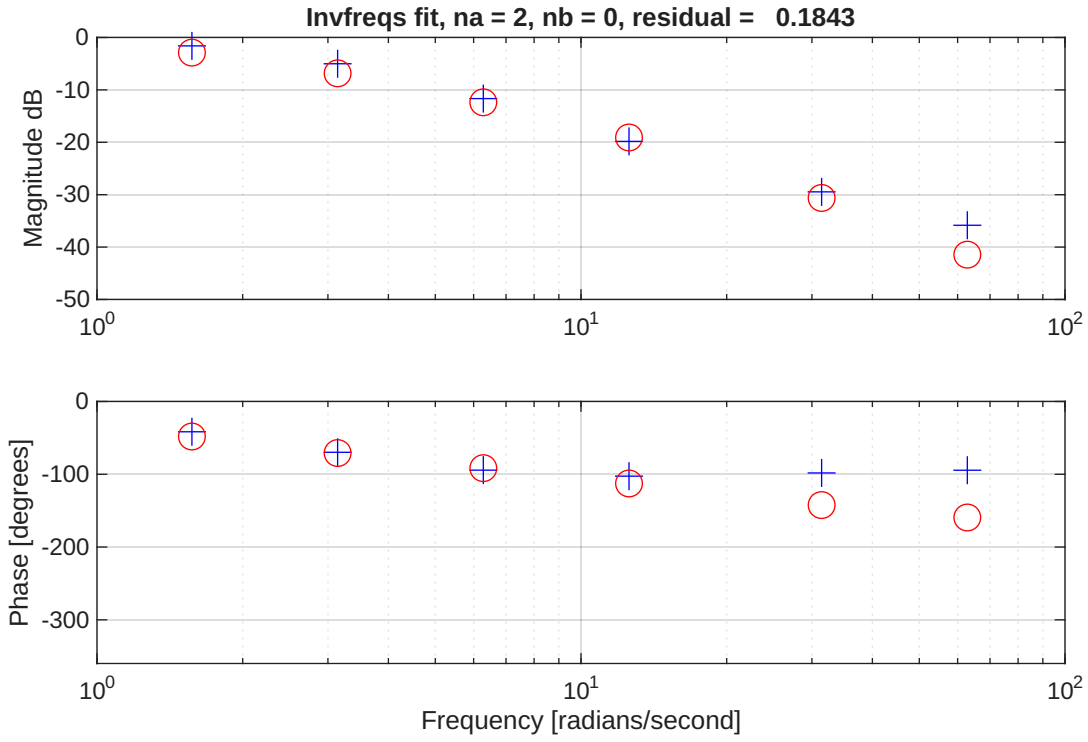


Figure 1: Frequency response data (+) and the 2-pole, 0-zero fit (o).

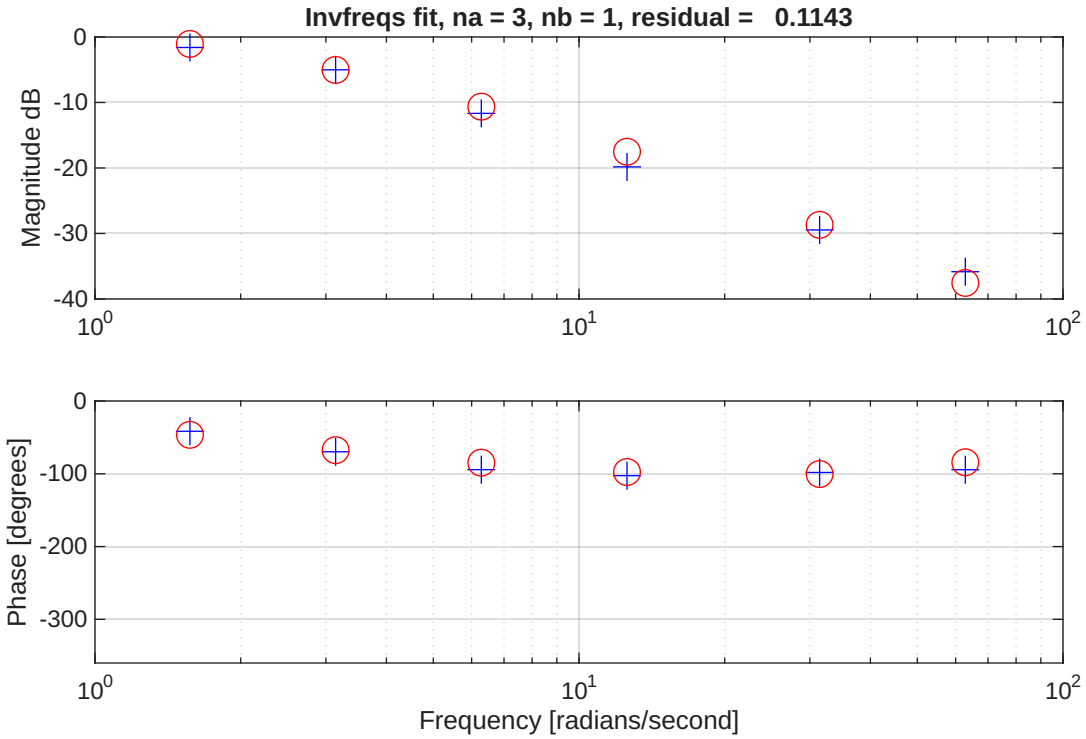


Figure 2: Frequency response data (+) and the 3-pole, 1-zero fit (o).

The transfer function with 3 poles and 1 zero provides a better fit.

Problem 1.B

Considering the pole-zero constellations, is each transfer function stable?

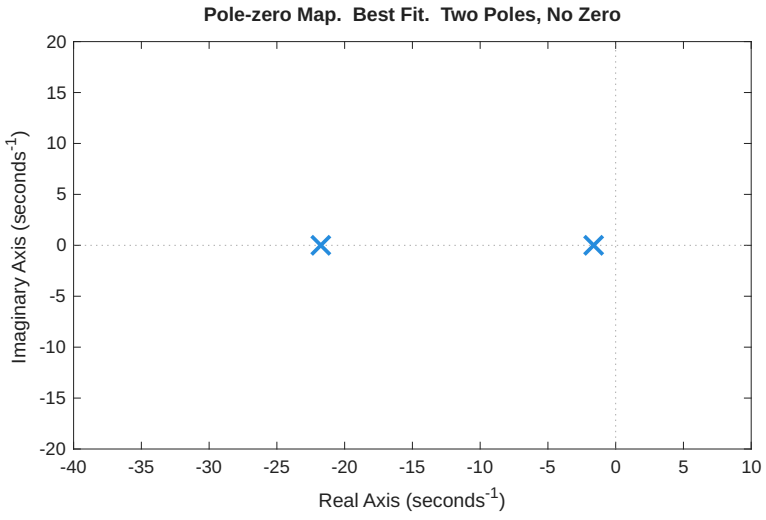


Figure 3: Pole-zero map of the 2-pole, 0-zero fit.

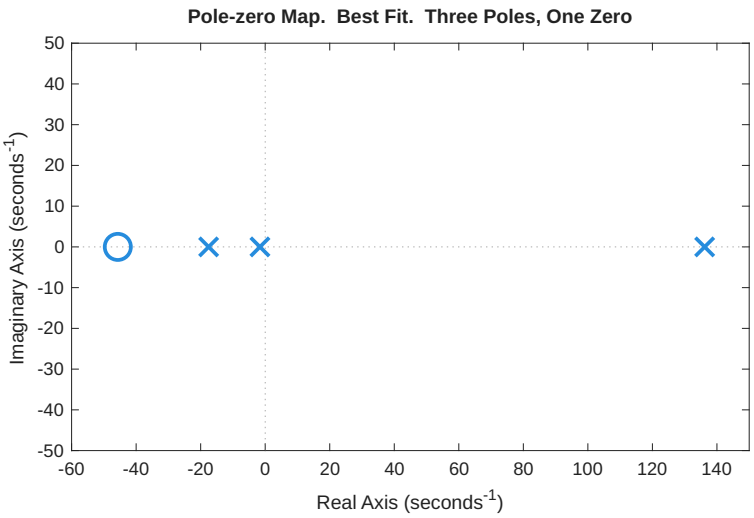


Figure 4: Pole-zero map of the 3-pole, 1-zero fit.

The transfer function with three poles and one zero is unstable.

Problem 1.C

Modify the data in `HW09_InvfreqsExample.m` *to be the following. This is the complex gain versus frequency for a system with transfer function* $G_{uy}(s) = B_b(s)/A_b(s)$.

Freq [rad/s]	Mag	Phase [deg]
0.3142	0.4180	-6.0062
0.6283	0.4237	-12.0334
1.2566	0.4426	-24.2801
3.1416	0.5135	-66.4484
6.2832	0.2910	-131.1225
12.5664	0.0754	-161.5733
31.4159	0.0120	-173.4103
62.8319	0.0030	-176.7621

Problem 1.C.1

Determine $G_{uy}(s)$ *for the models (indicate the transfer function):*

Model 1: 2 poles, 0 zeros

write the answer here

Model 2: 3 poles, 1 zero

write the answer here

Problem 1.C.2

Determine the better model based on these criteria:

Is the model physically realistic?

$G_{uy}(s)$ *is known to be open-loop stable — does the model have unstable poles? Does it have poles or zeros substantially faster than the fastest data point,* $\omega_{fastest} = 10\,Hz \cdot 2\pi = 62.8$ *rad/s? Is it physically infeasible in some other way? If so, reject the model.*

write the answer here

Goodness of fit

Does the model give a low residual?

write the answer here

Problem 2 — Control Design

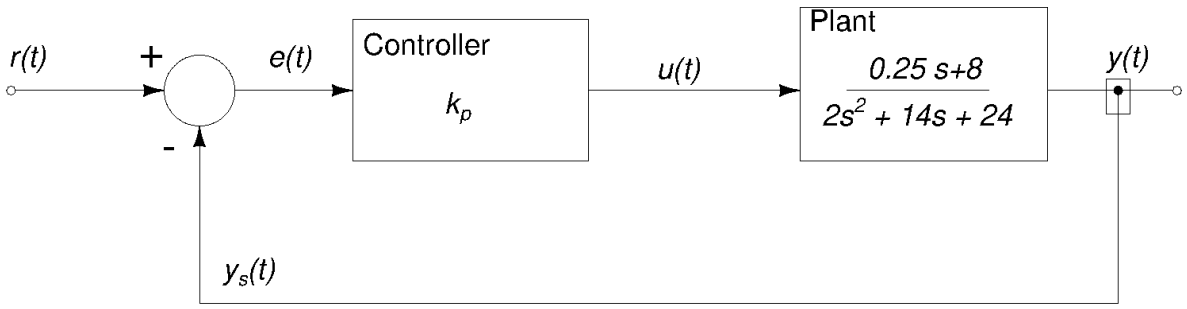


Figure 5: System with p-type control.

Test values of k_p on the range $k_p = 0.01 \ldots 100$. Select the value of k_p that best meets these design goals:

Measure	Goal
Rise time	Less than 1/2 second
Settling time	Less than 2 seconds
Steady-state error	As small as possible
Overshoot	Less than 10%

Steady-state error for a step input is defined as $e(t)$ after $y(t)$ reaches its steady value.

Problem 2.A

What is the suitable value of k_p?

write the answer here

Problem 2.B

As k_p increases, does rise time increase or decrease?

write the answer here

Problem 2.C

As k_p increases, does settling time increase or decrease?

write the answer here

Problem 2.D

As k_p increases, does steady-state error increase or decrease?

write the answer here

Problem 2.E

As k_p increases, does overshoot increase or decrease?

write the answer here